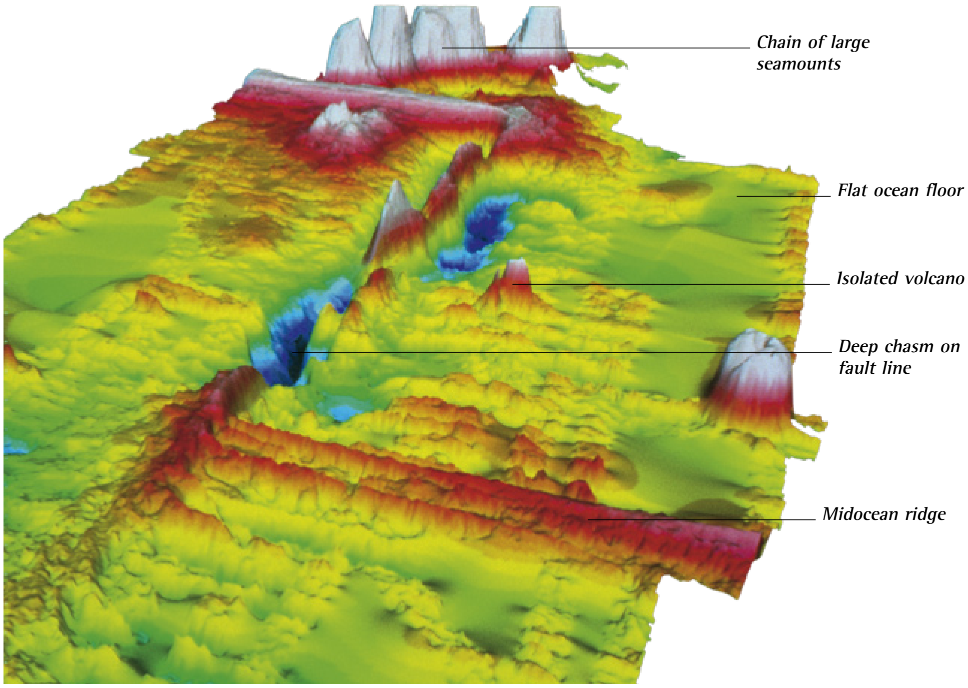
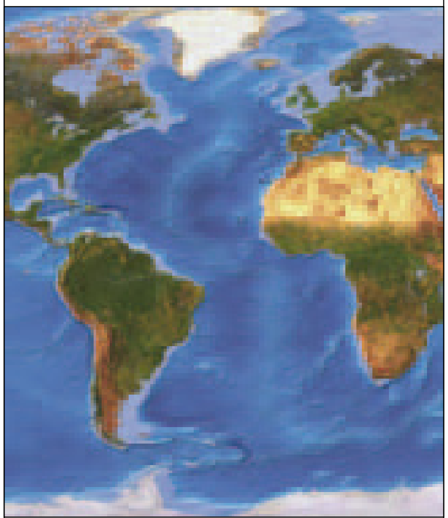
► SONAR SURVEYS
The early research ships spent a lot of time measuring ocean depths using extremely long, weighted sounding lines. This gave way to echo-sounding or sonar techniques, and eventually side-scan sonar, which produces 3-D views, as in this image of the Pacific ocean floor off South America.



◄ DEEP-SEA DRILLING
Many oceanic theories have been investigated by sampling the rocks of the ocean floor at various places and depths to build up a picture of its composition. The Japanese drilling vessel *Chikyu* can drill to depths of 23,000 ft (7,000 m) or more below the ocean floor, in waters 8,200 ft (2,500 m) deep—a total depth greater than the height of Mount Everest.

oceanography

SATELLITE INFORMATION



Data collected by satellites has proved extremely valuable to oceanographers. Gravity measurements taken by sensors mounted on satellites have provided the most accurate images yet of the ocean floors, as in this image of the Atlantic Ocean. Other sensors are able to gather data on temperatures, ice cover, ocean currents, and plankton distribution, which can then be used to build up accurate, up-to-date maps. Satellite images also give us superb views of oceanic weather systems, including hurricanes.